

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in the KDD process and justify the role of data preprocessing in it.

Definition

Data Mining is the process of extracting useful, hidden, and meaningful patterns or knowledge from large datasets.

Two Basic Data Mining Tasks

- Classification
- Clustering

KDD (Knowledge Discovery in Databases) Process

1. Data Cleaning
2. Data Integration
3. Data Selection
4. Data Transformation
5. Data Mining
6. Pattern Evaluation
7. Knowledge Presentation

Role of Data Preprocessing

- Removes missing and noisy data.
- Improves data quality.
- Increases mining accuracy.
- Reduces processing time.

Q2. What is Data Cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples.

Definition

Data Cleaning is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

Two Data Quality Issues

- Missing values
- Noisy or incorrect data

Handling Missing Values

- Ignore records
- Fill with mean, median, or mode

- Predict using machine learning

Example:

Ages = 20, 22, ?, 25

Mean = $(20+22+25)/3 = 22.33$

Replace missing value with 22.

Handling Noisy Data

- Binning
- Regression
- Clustering

Example:

Marks: 60, 61, 62, 95

Here 95 is noise (outlier).

Q3. Define Data Integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration.

Definition

Data Integration combines data from multiple sources into a single unified dataset.

Challenges

- Redundancy
- Data inconsistency

Handling Redundancy

- Remove duplicate records.
- Correlation analysis.

Handling Inconsistency

- Standardize formats.
- Resolve conflicting values using validation rules.

Example:

Database A: Male

Database B: M

Converted into a common format.

Q4. What is Covariance Analysis? List the two covariance formulas.

Definition

Covariance measures how two variables change together.

Interpretation

- Positive covariance → Variables increase together.
- Negative covariance → One increases while the other decreases.
- Zero covariance → No relationship.

Formulas

Population Covariance

$$Cov(Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{N}$$

Sample Covariance

$$Cov(Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Q5. Suppose two stocks A and B have values:

(2,5), (3,8), (5,10), (4,11), (6,14)

Determine whether the prices rise/fall together or independently.

Solution

A	B
2	5
3	8
5	10
4	11
6	14

Mean(A)=4

Mean(B)=9.6

$$Cov = \frac{26}{4} = 6.5$$

Since covariance is positive, the stock prices rise and fall together.

Answer: The two stocks are positively related and move together.

Q6. Explain the architecture of a typical data mining system.

Components

1. Data Sources

- Databases, files, web.

2. Database/Data Warehouse

- Stores integrated data.

3. Data Cleaning & Integration

- Removes errors and combines data.

4. Data Mining Engine

- Performs classification, clustering, association, prediction.

5. Pattern Evaluation

- Identifies useful patterns.

6. Knowledge Presentation

- Displays graphs, reports, dashboards.

Interaction

All components work together to convert raw data into useful knowledge for decision-making.

Q7. Find covariance

Given

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

Solution

Mean(x)

$$= \frac{4+5+6+7+9}{5} = 6.2$$

Mean(y)

$$= \frac{8+3+7+5+6}{5} = 5.8$$

x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	Product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64

x	y	(x- $\bar{x}$)	(y- $\bar{y}$)	Product
9	6	2.8	0.2	0.56
Sum = -1.80				
Sample covariance				
$= \frac{-1.80}{4} = -0.45$				

Answer

Covariance = -0.45

Interpretation:

The variables have a weak negative relationship.

Q8. Describe the major steps in Data Preprocessing. Propose a preprocessing pipeline.

Major Steps

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization

Pipeline

Raw Data

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Data Cleaning

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Data Integration

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Data Transformation

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Data Reduction

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Data Mining

Justification

Cleaning first removes errors.

Integration combines sources.

Transformation normalizes data.

Reduction decreases dataset size.

Finally, mining becomes faster and more accurate.

Q9. Hospital Age Data – Equal-Frequency (Equi-Depth) Binning

Data:

13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70

There are 27 values.

Using 3 equal-frequency bins

Each bin contains 9 values.

Bin 1

13,15,16,16,19,20,20,21,22

Mean = 18

Bin 2

22,25,25,25,25,30,33,33,35

Mean = 28

Bin 3

35,35,35,36,40,45,46,52,70

Mean = 43

Smoothed Data

18 18 18 18 18 18 18 18 18

28 28 28 28 28 28 28 28 28

43 43 43 43 43 43 43 43 43

This reduces noise while preserving the overall trend.

Q10. Explain Data Discretization and compare Equal-Width and Equal-Frequency Binning.

Definition

Data discretization converts continuous values into discrete intervals (bins).

Importance

- Reduces complexity.
- Removes noise.
- Improves mining efficiency.
- Makes data easier to understand.

Equal-Width Binning

Formula

$$Width = \frac{Maximum - Minimum}{Number\ of\ bins}$$

Example

Marks:

10,20,30,40,50,60

Bins

10–30

30–50

50–70

Advantages

- Simple.
- Fast.

Disadvantages

- Uneven number of records in each bin.

Equal-Frequency (Equi-Depth) Binning

Each bin contains the same number of records.

Example

Data:

10,20,30,40,50,60

Bins

(10,20)

(30,40)

(50,60)

Advantages

- Balanced bins.

- Better for skewed data.

Disadvantages

- Bin widths vary.

Comparison

Feature	Equal-Width	Equal-Frequency
Bin Size	Fixed width	Fixed number of records
Bin Width	Same	Different
Distribution	May be uneven	Balanced
Best For	Uniform data	Skewed data